

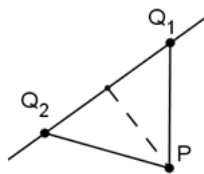
Qn	Solutions	Comments
1i	$\frac{dh}{dt} = \frac{1}{10} \sqrt{16 - \frac{h}{2}}$ At maximum height, $\frac{dh}{dt} = 0$ $16 - \frac{h}{2} = 0 \quad \Leftrightarrow \quad h = 32$ Maximum height = 32m	
ii	$\frac{dh}{dt} = \frac{1}{10} \sqrt{16 - \frac{h}{2}}$ $\int \left(16 - \frac{h}{2}\right)^{-\frac{1}{2}} dh = \int \frac{1}{10} dt$ $\frac{\left(16 - \frac{h}{2}\right)^{\frac{1}{2}}}{-\frac{1}{2}\left(\frac{1}{2}\right)} = \frac{t}{10} + C$ $-4\left(16 - \frac{h}{2}\right)^{\frac{1}{2}} = \frac{t}{10} + C$ $t = D - 40\left(16 - \frac{h}{2}\right)^{\frac{1}{2}}$ When $t = 0$, $h = 0$ $0 = D - 40(16)^{\frac{1}{2}} \Rightarrow D = 160$ $t = 160 - 40\left(16 - \frac{h}{2}\right)^{\frac{1}{2}}$ When $h = 16$, $t = 160 - 40(16 - 8)^{\frac{1}{2}} = 46.9$ The tree takes 46.9 years to grow to half the maximum height.	By writing $\frac{1}{\sqrt{a+bh}}$ as $(a+bh)^{-\frac{1}{2}}$, it becomes easier to integrate. Do not mistake it for $\frac{f'(h)}{f(h)}$. If $f(h) = \sqrt{16 - \frac{h}{2}}$, we will need to see $f'(h) = \frac{1}{2\sqrt{16 - \frac{h}{2}}} \left(-\frac{1}{2}\right)$
2i	Equation of line L is $\vec{r} = \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}, \lambda \in \mathbb{R}$ Let θ be the angle between L and the x-axis $\cos \theta = \frac{\begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}}{\sqrt{49}\sqrt{1}} = \frac{2}{7}$ $\theta = 73.4^\circ$	Do note that the x-axis is a line with equation $\vec{r} = \mu \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \mu \in \mathbb{R}$

ii

Let Q be a point on L .

$$\overrightarrow{OQ} = \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}, \text{ for some } \lambda.$$

$$\begin{aligned} \overrightarrow{PQ} &= \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix} - \begin{pmatrix} 2 \\ 5 \\ -6 \end{pmatrix} \\ &= \begin{pmatrix} -1+2\lambda \\ -7+3\lambda \\ 2-6\lambda \end{pmatrix} \end{aligned}$$



$$\begin{aligned} PQ &= \sqrt{(-1+2\lambda)^2 + (-7+3\lambda)^2 + (2-6\lambda)^2} \\ &= \sqrt{54 - 70\lambda + 49\lambda^2} \end{aligned}$$

$$54 - 70\lambda + 49\lambda^2 = 33$$

$$49\lambda^2 - 70\lambda + 21 = 0$$

$$7\lambda^2 - 10\lambda + 3 = 0$$

$$\lambda = 1 \text{ or } \frac{3}{7}$$

$$\overrightarrow{OQ} = \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ -10 \end{pmatrix} \text{ or}$$

$$\overrightarrow{OQ} = \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} + \frac{3}{7} \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix} = \frac{1}{7} \begin{pmatrix} 13 \\ -5 \\ -46 \end{pmatrix}$$

The position vector of the points on L which are $\sqrt{33}$ from P

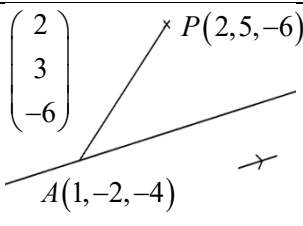
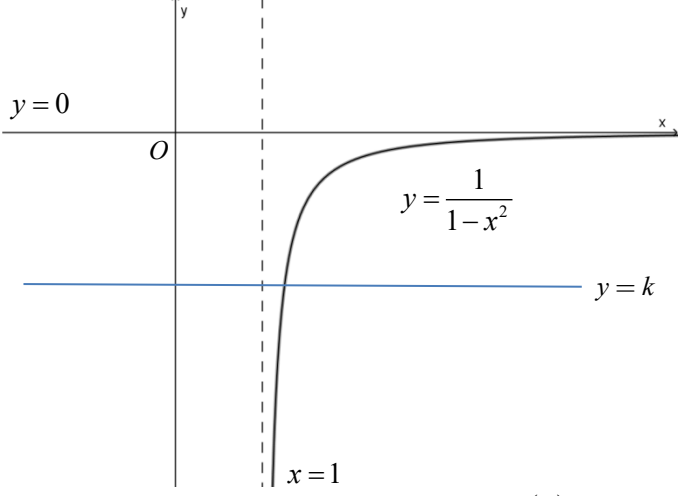
$$\text{are } \begin{pmatrix} 3 \\ 1 \\ -10 \end{pmatrix} \text{ and } \frac{1}{7} \begin{pmatrix} 13 \\ -5 \\ -46 \end{pmatrix}$$

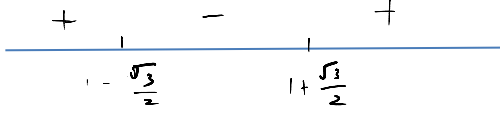
Point on L closest to P is the mid point of the 2 possible points for Q .

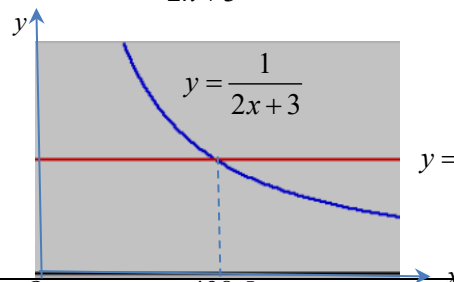
$$\text{Mid point is } \frac{\begin{pmatrix} 3 \\ 1 \\ -10 \end{pmatrix} + \frac{1}{7} \begin{pmatrix} 13 \\ -5 \\ -46 \end{pmatrix}}{2} = \frac{1}{7} \begin{pmatrix} 17 \\ 1 \\ -58 \end{pmatrix}$$

$$\frac{1}{7} \begin{pmatrix} 17 \\ 1 \\ -58 \end{pmatrix} \text{ is the position vector of point on } L \text{ closest to } P.$$

Think of an isosceles triangle.

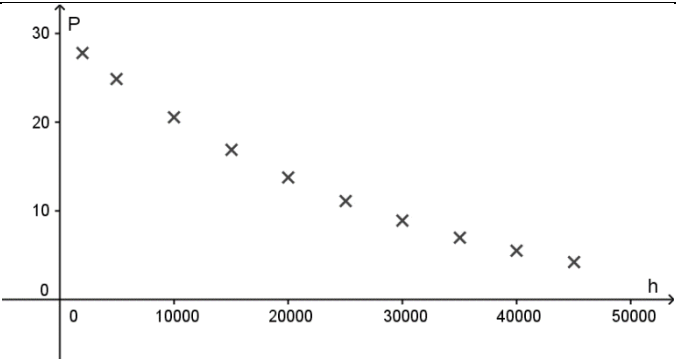
iii	$\overrightarrow{AP} = \begin{pmatrix} 2 \\ 5 \\ -6 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \\ -2 \end{pmatrix}$ $\vec{n} = \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix} \times \begin{pmatrix} 1 \\ 7 \\ -2 \end{pmatrix} = \begin{pmatrix} 36 \\ -2 \\ 11 \end{pmatrix}$ $\begin{pmatrix} 1 \\ -2 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 36 \\ -2 \\ 11 \end{pmatrix} = -4$ Equation of plane is $36x - 2y + 11z = -4$ 	
3ai	 Every horizontal line cuts the graph of $y = f(x)$ at most once. $\therefore f$ is 1-1 and f^{-1} exists.	
ii	Let $y = \frac{1}{1-x^2}$ $y - x^2 y = 1$ $x^2 = \frac{y-1}{y} = 1 - \frac{1}{y}$ $x = \pm \sqrt{1 - \frac{1}{y}} \quad (\text{reject } -ve \because x > 1)$ $f^{-1}(x) = \sqrt{1 - \frac{1}{x}}$ $D_{f^{-1}} = (-\infty, 0)$	
b	Let $y = \frac{2+x}{1-x^2}$ $y - x^2 y = 2 + x$ $yx^2 - x + 2 - y = 0$ For real values of x, $(-1)^2 - 4(y)(2-y) \geq 0$	Since the question mentioned “algebraically”, graphical method is not allowed. The idea is that for y to be in the range, there needs to be some real values of x to yield that value of y. Hence we use the discriminant of

	$4y^2 - 8y + 1 \geq 0$ $y^2 - 2y + \frac{1}{4} \geq 0$ $(y-1)^2 - \frac{3}{4} \geq 0$ $\left(y-1+\frac{\sqrt{3}}{2}\right)\left(y-1-\frac{\sqrt{3}}{2}\right) \geq 0$  $R_g = \left(-\infty, 1 - \frac{\sqrt{3}}{2}\right] \cup \left[1 + \frac{\sqrt{3}}{2}, \infty\right)$	the quadratic equation in x to obtain an inequality for y.
4a	Let P_n be the statement $\sum_{r=1}^n r(r+2)(r+5) = \frac{1}{12}n(n+1)(3n^2 + 31n + 74), \quad n \in \mathbb{Z}^+$ Check P_1: LHS = $1(3)(6) = 18$ RHS = $\frac{1}{12}1(2)(3 + 31 + 74) = 18$ $\therefore P_1$ is true. Assume the P_k is true for some positive integer k, i.e. $\sum_{r=1}^k r(r+2)(r+5) = \frac{1}{12}k(k+1)(3k^2 + 31k + 74)$ We want to show P_{k+1} is true, i.e. $\begin{aligned} & \sum_{r=1}^{k+1} r(r+2)(r+5) \\ &= \frac{1}{12}(k+1)(k+2)(3(k+1)^2 + 31(k+1) + 74) \\ &= \frac{1}{12}(k+1)(k+2)(3k^2 + 37k + 108) \end{aligned}$ LHS = $\sum_{r=1}^{k+1} r(r+2)(r+5)$ $\begin{aligned} &= \frac{1}{12}k(k+1)(3k^2 + 31k + 74) + (k+1)(k+3)(k+6) \\ &= \frac{1}{12}(k+1)(3k^3 + 31k^2 + 74k + 12k^2 + 108k + 216) \\ &= \frac{1}{12}(k+1)(3k^3 + 43k^2 + 182k + 216) \\ &= \frac{1}{12}(k+1)(k+2)(3k^2 + 37k + 108) \\ &= \text{RHS} \end{aligned}$	Not in the H2 Math 9758 syllabus It is not compulsory to write the expression as a summation but it is something you would be more used to seeing. The last term in the series “$n(n+2)(n+5)$” gives us a hint on the what the general term should look like. Be careful of your statements. Do not be too eager to expand, factorise where possible.

	$\therefore P_{k+1}$ is true. Since P_1 is true and P_k is true $\Rightarrow P_{k+1}$ is true By mathematical induction, P_n is true for all positive integers n.	
bi	$\frac{2}{4r^2+8r+3} = \frac{2}{(2r+1)(2r+3)} = \frac{A}{2r+1} + \frac{B}{2r+3}$ By Cover up rule $A = \frac{2}{2\left(-\frac{1}{2}\right)+3} = 1, \quad B = \frac{2}{2\left(-\frac{3}{2}\right)+1} = -1$ $\frac{2}{4r^2+8r+3} = \frac{1}{2r+1} - \frac{1}{2r+3}$	
	$S_n = \sum_{r=1}^n \frac{2}{4r^2+8r+3}$ $= \sum_{r=1}^n \left(\frac{1}{2r+1} - \frac{1}{2r+3} \right)$ $= \frac{1}{3} - \frac{1}{5} + \frac{1}{5} - \frac{1}{7} + \dots + \frac{1}{2n-1} - \frac{1}{2n+1} + \frac{1}{2n+1} - \frac{1}{2n+3}$ $= \frac{1}{3} - \frac{1}{2n+3}$	
ii	Sum to infinity = $\frac{1}{3}$ $\left \frac{1}{3} - \left(\frac{1}{3} - \frac{1}{2n+3} \right) \right < 10^{-3}$ $\left \frac{1}{2n+3} \right < 10^{-3}$ $-10^{-3} < \frac{1}{2n+3} < 10^{-3}$ Since $n > 0$, $0 < \frac{1}{2n+3} < 10^{-3}$ 	

	From the graph, $n > 498.5$ Smallest value of n is 499.	
5i	It will be difficult to get the sampling frame, which is a list of all the customers of the supermarket.	Not in the H2 Math 9758 syllabus
ii	He can split the customers into strata: customers below 20 years old, customers from 20 to 40 years old and customers more than 40 years old. He surveys the number of customers required for each strata by approaching customers that wishes to approach.	Not in the H2 Math 9758 syllabus
iii	He may end up with a bias sample, e.g. female customers only.	Not in the H2 Math 9758 syllabus
6i	Let R be the number of red sweets, in a packet of 10. $R \sim B(10, 0.25)$ $P(R \geq 4) = 1 - P(R \leq 3)$ $= 0.224$	
ii	Let S be the number of red sweets, in a packet of 100 $S \sim B(100, 0.25)$ Since n is large, $np = 25 > 5$ and $n(1 - p) = 75 > 5$ $S \sim N(25, 18.75)$ approximately $P(S \geq 30) = P(S \geq 29.5)$ $= 0.14938$ $= 0.149$	Not in the H2 Math 9758 syllabus
iii	Let T be the number of large packets of Droppers that contain at least 30 red sweets, out of 15 $T \sim B(15, 0.14938)$ $P(S \leq 3) = 0.825$	
7i	The assumptions are errors occur independently, and the average number of errors per page is a constant.	Not in the H2 Math 9758 syllabus Be careful of your phrasing here. We do not say things like the number of errors are independent. Neither do we say average number of errors is a constant. Need to bring across some idea of average rate.
ii	Let X be the number of errors on the first 6 pages $X \sim \text{Po}(7.8)$ $P(X > 10) = 1 - P(X \leq 10)$ $= 0.165$	Not in the H2 Math 9758 syllabus
iii	Let Y be the number of errors on the first n pages	Not in the H2 Math 9758 syllabus

	$Y \sim \text{Po}(1.3n)$ $P(Y \leq 1) < 0.05$ $e^{-1.3n} \left(\frac{(1.3n)^0}{0!} + \frac{(1.3n)^1}{1!} \right) < 0.05$ $e^{-1.3n} (1 + 1.3n) < 0.05$ From GC <table border="1"><tr><td>n</td><td>$e^{-1.3n} (1 + 1.3n)$</td></tr><tr><td>3</td><td>0.0992</td></tr><tr><td>4</td><td>0.034</td></tr></table> Hence, the least value of n is 4.	n	$e^{-1.3n} (1 + 1.3n)$	3	0.0992	4	0.034	Since n is an integer value, we can solve the inequality, we could solve the inequality by looking at the table of values.
n	$e^{-1.3n} (1 + 1.3n)$							
3	0.0992							
4	0.034							
8	By GC, Unbiased estimate of population mean = 0.8825 Unbiased estimate of population variance = 0.0055929 = 0.00559 Let X kg be the mass of a randomly chosen pineapple, and μ kg be the population mean mass. $H_0: \mu = 0.9$ $H_1: \mu < 0.9$ Level of significance: 10% Test statistics: When H_0 is true, $T = \frac{\bar{X} - 0.9}{S/\sqrt{8}}$ Computation: $\nu = 7$, $p\text{-value} = 0.265$ Since $p\text{-value} = 0.265 > 0.10$, there is insufficient evidence to reject H_0 at 10% level of significance. Hence, there is insufficient evidence to doubt the owner's claim at 10% level of significance.	Not in the H2 Math 9758 syllabus Since this is a one-tailed test and $\bar{x} < 0.9$, hence we choose $H_1: \mu < 0.9$. The other way to view this question is that we are testing for evidence to doubt the claim. If there is evidence to doubt the claim, then $\mu < 0.9$. Hence we choose $H_1: \mu < 0.9$.						
9i	Since events A and B are independent, $P(B A) = P(B) = 0.4$							
ii	$P(A \cup B \cup C)$ $= P(A) + P(B) + P(C)$ $- P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$ $= 0.45 + 0.4 + 0.3 - (0.45)(0.4)$ $- (0.45)(0.3) - (0.4)(0.3) + 0.1$ $= 0.815$ $P(A' \cap B' \cap C') = 1 - 0.815 = 0.185$	Use a Venn diagram to visualize where $A' \cap B' \cap C'$ should be						

iii	From Venn Diagram, smallest $x = 0$ Largest $P(A \cup B \cup C) = 0.45 + 0.22 + 0.165 = 0.835$ Smallest $P(A' \cap B' \cap C') = 1 - 0.835 = 0.165$ From Venn Diagram, largest $x = 0.165$ Smallest $P(A \cup B \cup C) = 0.45 + (0.22 - 0.165) + 0.165 = 0.67$ Largest $P(A' \cap B' \cap C') = 1 - 0.67 = 0.33$	If events B and C are not independent, $P(B \cap C)$ is not $P(B)P(C)$. Hence, our aim is the to find the max and min values of $P(B \cap C)$ (or the value of x from our Venn Diagram) This is actually a bit similar to one of the practice questions in the summary revision package.
10i		
11a	$r = -0.9807$	
b	$r = -0.9748$	
c	$r = -0.9986$	
iii	By GC, $P = -0.14687\sqrt{h} + 34.789$ $P = -0.147\sqrt{h} + 34.8$	
iv	If h is in metres, $P = -0.14687\sqrt{3.28h} + 34.789$ $= -0.266\sqrt{h} + 34.8$	
11i	C A B G E S A B Number of ways = $\frac{8!}{2!2!} = 10080$	
ii	Number of arrangements = $10080 - 1 = 10079$	There is only 1 way to arrange the letters in alphabetical order
iii	Number of ways = $6! = 720$	Group AA and BB together.
iv	Number of ways with both As together $= \frac{7!}{2!} = 2520$ Number of ways with both Bs together $= \frac{7!}{2!} = 2520$ Number of ways = $10080 - (2(2520) - 720) = 5760$	This part uses the complement method and $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ We do not use separation method for this question. We need both the As and both the Bs. However, we do need to separate A from B.

12i	Let A_g and P_g be the mass of a randomly chosen apple and a randomly chosen pear respectively. Let $X = A_1 + A_2 + \dots + A_5$ $E(X) = 5E(A) = 1500$ $\text{Var}(X) = 5\text{Var}(A) = 2000$ $X \sim N(1500, 2000)$ $P(X > 1600) = 0.0127$	
ii	Let $Y = P_1 + P_2 + \dots + P_8$ $E(X - Y) = E(X) - 8E(P) = 1500 - 1600 = -100$ $\text{Var}(X - Y) = \text{Var}(X) + 8\text{Var}(P) = 2000 + 1800 = 3800$ $X - Y \sim N(-100, 3800)$ $P(X - Y > 0) = 0.0524$	
iii	Let $W = 0.85X + 0.9Y$ $E(W) = 0.85E(X) + 0.9E(Y)$ $= 0.85(1500) + 0.9(1600)$ $= 2715$ $\text{Var}(W) = (0.85)^2 \text{Var}(X) + (0.9)^2 \text{Var}(Y)$ $= (0.85)^2 (2000) + (0.9)^2 (1800)$ $= 2903$ $W \sim N(2715, 2903)$ $P(W < 2750) = 0.742$	

Note

Q8 was marked as “out of syllabus” in the students’ copy. Please modify the question to the suggested one below and get them to attempt on the spot.

- (8) A market sells pineapples which have masses that are normally distributed. The stall owner claims that the mean mass of the pineapples is at least 0.9kg.
- (i) Nur buys a random selection of 8 pineapples from the stall. The 8 pineapples have masses, in kg, as follows.
0.80 1.00 0.82 0.85 0.93 0.96 0.81 0.89
Find unbiased estimates of the population mean and variance of the mass of pineapples.
- (ii) A second random sample of 80 pineapples is bought from the stall. The unbiased estimates of the population mean and variance of the mass of pineapples are found to be the same as in part (i). Test at the 10% level of significance whether there is any evidence to doubt the stall owner’s claim.

8i	By GC, Unbiased estimate of population mean = 0.8825 Unbiased estimate of population variance = 0.0055929 = 0.00559	
ii	Let X kg be the mass of a randomly chosen pineapple, and μ kg be the population mean mass. $H_0: \mu = 0.9$ $H_1: \mu < 0.9$ Level of significance: 10% Test statistics: As $n = 80$ is large, by Central Limit Theorem, $\bar{X}$ is approximately normal. When H_0 is true, $Z = \frac{\bar{X} - 0.9}{S / \sqrt{80}} \sim N(0,1)$ approximately Computation: p-value = 0.0181 Since p-value = 0.0181 $<$ 0.10, H_0 is rejected at 10% level of significance. Hence, there is sufficient evidence to doubt the owner's claim at 10% level of significance.	We are testing for evidence to doubt the claim. If there is evidence to doubt the claim, then $\mu < 0.9$. Hence we choose $H_1: \mu < 0.9$. Conclusion needs to be contextualised